



**ATAL BIHARI VAJPAYEE GOVERNMENT INSTITUTE OF
ENGINEERING & TECHNOLOGY**

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INDUSTRIAL INTERNSHIP REPORT

on

CROP AND WEED DETECTION USING YOLO

Prepared by

Ankit Thakur

Department of Computer Science & Engineering

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Executive Summary

This report presents the work completed during the industrial internship on the project “Crop and Weed Detection Using YOLO”. The project focuses on applying computer vision and deep learning to identify agricultural objects in images, with particular emphasis on distinguishing crop plants from weeds. Accurate detection of weeds is important because unwanted plants compete with crops for water, nutrients, sunlight and growing space.

The proposed system uses the YOLO (You Only Look Once) family of object-detection models through the Ultralytics framework. Unlike image-classification systems that assign one label to an entire image, an object detector predicts the class and location of individual objects using bounding boxes. This makes the approach suitable for practical agricultural scenes where crop and weed plants may occur together.

The internship work involved understanding the problem, preparing the dataset, configuring the YOLO training environment, preprocessing and organizing data, training the detection model, monitoring training behaviour, validating predictions, and analysing the limitations of the developed solution. The implementation was carried out in Python using commonly used computer-vision and machine-learning libraries including Ultralytics YOLO, OpenCV, NumPy, Pandas, Matplotlib and PyYAML.

The project provided practical exposure to the complete machine-learning workflow rather than only model training. It also highlighted real-world issues such as dataset quality, class imbalance, annotation quality, hardware limitations, Python/package compatibility and the trade-off between detection accuracy and inference speed.

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1 Preface

The industrial internship provided an opportunity to move beyond classroom concepts and work on a practical computer-vision problem. The internship project, “Crop and Weed Detection Using YOLO”, was selected because agriculture is an important application area for artificial intelligence and because automated weed detection can support precision-agriculture systems.

During the internship, the work progressed through several stages: understanding the problem statement, studying object detection, preparing the development environment, organizing the dataset, training the YOLO model, observing training metrics, testing predictions and documenting the complete workflow.

The project also demonstrated that a machine-learning system is strongly dependent on the quality and diversity of its training data. Variations in illumination, camera angle, plant size, soil background, occlusion and growth stage can affect detection performance. Therefore, model development was treated as an iterative process rather than simply running a training command.

The internship improved my practical understanding of Python, computer vision, deep learning, dataset preparation, model training, debugging and technical documentation. It also helped me understand how an academic ML model can be translated into a system with a potential real-world use case.

I express my sincere gratitude to the internship coordinators, mentors, faculty members and everyone who supported the completion of this work. I also thank my institution for providing the opportunity to gain practical exposure through the internship.

For juniors and peers, the main lesson from this project is to spend sufficient time understanding the dataset and problem before tuning the model. A technically advanced model cannot compensate for poor labels, weak data or an unclear evaluation strategy.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is an industrial technology organization working in the digital transformation domain, with solutions involving technologies such as Internet of Things, cloud computing, machine learning, communication technologies, software development and industrial monitoring. The internship format provided exposure to an industry-oriented problem and required the project and documentation to be completed within the internship schedule.

UCT's technology focus includes IoT platforms, smart-factory solutions, industrial monitoring, agritech and predictive-maintenance applications. These domains demonstrate how software, sensors, data analytics and machine learning can be combined to solve operational problems.

Technology Relevance to the Internship

- Internet of Things enables acquisition of data from physical environments and devices.
- Cloud and software platforms provide scalable infrastructure for data processing and application deployment.
- Machine learning converts collected data into predictions, classifications or detections.
- Computer vision extends this approach to image and video data.
- Agritech applications can use computer vision to monitor crops, identify weeds and support precision farming.

The crop-and-weed detection project fits this broader technology landscape because a camera-based detection system can act as an input component for future precision-agriculture or smart-farming applications.

2.2 About upskill Campus and The IoT Academy

The internship program was facilitated through the training and internship ecosystem associated with upskill Campus and The IoT Academy in collaboration with the industrial partner. The program was intended to provide students with exposure to industrial problem solving, project implementation and professional skills.

2.3 Objectives of the Internship

- Gain practical experience in solving an industry-oriented technical problem.
- Understand the lifecycle of a machine-learning project from data preparation to evaluation.
- Develop practical skills in Python and computer-vision libraries.
- Understand object detection and YOLO-based model training.
- Learn to troubleshoot development-environment and dependency issues.
- Improve technical documentation, analysis and communication skills.
- Understand the constraints that affect deployment of AI models in real-world environments.

Internship Learning Outcome

The internship converted theoretical knowledge of artificial intelligence and machine learning into an applied workflow. In particular, the project required understanding the relationship between data quality, model configuration, training behaviour and detection results.

2.4 Project Objectives

1. Develop a computer-vision system capable of detecting crop and weed instances in agricultural images.
2. Use a YOLO-based object-detection model to predict object classes and bounding boxes.
3. Prepare and structure the dataset in a format suitable for YOLO training.
4. Train and validate the model using a reproducible Python environment.
5. Evaluate the trained model using qualitative predictions and standard object-detection metrics where available.
6. Create a foundation that can later be extended to real-time camera/video-based agricultural monitoring.

2.5 References

7. Ultralytics, YOLO documentation and training framework, used as the implementation reference for YOLO-based object detection.
8. J. Redmon, S. Divvala, R. Girshick and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” Proceedings of CVPR, 2016.
9. G. Jocher et al., Ultralytics YOLO ecosystem and associated documentation for model training and inference.
10. OpenCV documentation, used as a reference for image-processing and computer-vision operations.
11. NumPy, Pandas and Matplotlib documentation for numerical processing, dataset handling and visualization.

2.6 Glossary

Term	Meaning
YOLO	You Only Look Once; a family of real-time object-detection models.
Object Detection	Task of identifying objects and locating them using bounding boxes.
Bounding Box	Rectangular region indicating the location of a detected object.
Class	Category assigned to an object, such as crop or weed.
Epoch	One complete pass through the training dataset.
Precision	Fraction of predicted positive detections that are correct.
Recall	Fraction of relevant objects that are successfully detected.
mAP	Mean Average Precision, a common object-detection evaluation metric.
IoU	Intersection over Union; overlap measure between predicted and ground-truth boxes.

3 Problem Statement

Manual identification of weeds in agricultural fields is time-consuming and depends on human observation. When weeds are distributed among crops, identifying their exact locations can be difficult, especially when plants are small, partially occluded or visually similar.

The problem addressed in this internship is to develop an automated image-based system that can locate and classify crop and weed objects using deep-learning-based object detection. The system should accept an image containing agricultural vegetation and produce bounding boxes with class predictions for the detected objects.

Problem Requirements

- Input should be an agricultural image containing one or more target objects.
- The model should identify the target classes defined by the dataset.
- Each detection should include a bounding box and confidence score.
- The system should be computationally practical enough to support future real-time or near-real-time use.
- The training workflow should be reproducible and should allow the model to be retrained when more data become available.

Practical Challenges

- Crop and weed appearance changes with growth stage and environmental conditions.
- Leaves can overlap, causing partial occlusion.
- Lighting and shadows can change image appearance.
- Background soil and other vegetation may resemble target objects.
- Small objects are harder to detect than large, clearly visible objects.
- Model performance depends heavily on annotation quality and dataset diversity.

4 Existing and Proposed Solution

4.1 Existing Approaches and Limitations

Traditional weed-management practices often rely on manual scouting, mechanical removal or broad application of herbicides. These methods can be effective but do not provide precise, image-level information about where weeds are located.

Earlier computer-vision approaches may use handcrafted features such as colour, texture or shape followed by classical machine-learning algorithms. Such methods can work in controlled conditions but can become less robust when lighting, soil background, plant size or growth stage changes.

Image-classification networks can determine whether a particular image contains a class, but classification alone does not provide the exact location of each crop or weed. Object detection is therefore more appropriate when the system must locate multiple objects in the same image.

Limitations of Existing Approaches

- Manual inspection is labour intensive and difficult to scale.
- Rule-based image processing can be sensitive to environmental conditions.
- Image classification does not directly provide object locations.
- High-accuracy detection models can require substantial computational resources.
- A model trained on a narrow dataset may not generalize well to different fields or cameras.

4.2 Proposed Solution

The proposed solution uses a YOLO-based object detector to identify crop and weed instances directly from images. The system is designed around a dataset-driven training workflow: annotated images are prepared in the required format, the YOLO model is trained on the dataset, validation is performed during training, and the resulting weights are used for inference on unseen images.

The implementation uses the Ultralytics framework because it provides a practical interface for model training, validation and prediction. Python acts as the primary programming language, while OpenCV can be used for image handling and visualization, NumPy for numerical operations, Pandas for tabular data handling, Matplotlib for plots and PyYAML for dataset/model configuration.

Value Addition

- Automated localization of crops and weeds.
- Potential reduction in manual scouting effort.
- Foundation for targeted weed-management systems.
- Possibility of extension from still images to video streams.
- Potential integration with drones, field cameras or agricultural robots in future work.
- Reusable training pipeline that can be improved using additional annotated field data.

4.3 Code and Report Submission

Code repository: To be inserted with the final project repository URL.

Final report repository: To be inserted if the internship requires online report submission.

Model weights and dataset configuration: To be retained with the project files for reproducibility.

5 Proposed Design / Model

The overall design follows a standard object-detection pipeline. First, the input dataset is collected and annotated. The images and labels are organized into training and validation subsets. The YOLO configuration specifies the dataset location and target classes. During training, the model learns visual features and object locations from the annotated examples. After training, the learned weights are used to perform inference on new images.

5.1 High-Level Architecture

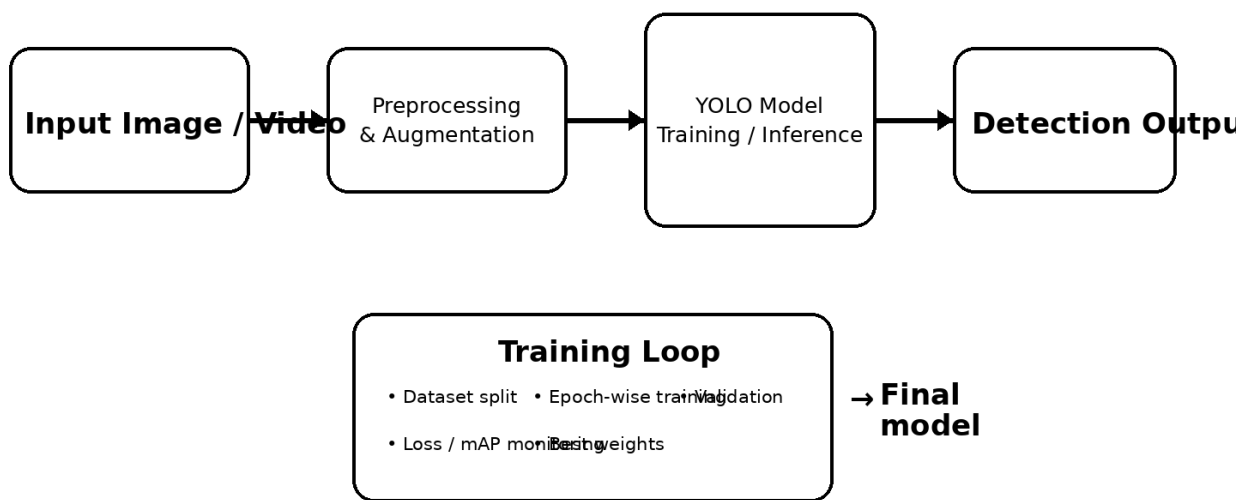


Figure 1: High-level architecture of the crop and weed detection system

The architecture separates data preparation, model training and inference. The same trained model can be applied to images that were not included in training, which is essential for evaluating generalization.

5.2 Dataset and Preprocessing

Object-detection training requires images together with annotations that identify the location and class of each target object. For YOLO training, annotations are represented using normalized bounding-box coordinates and class identifiers. The dataset configuration also specifies the class names and the locations of the training and validation data.

Preprocessing Workflow

12. Collect and inspect agricultural images.
13. Verify that each target object has an appropriate bounding-box annotation.
14. Organize images and label files according to the YOLO dataset structure.
15. Define the target classes, such as crop and weed, in the dataset configuration.
16. Apply suitable image transformations/augmentation during training where supported.
17. Check the training and validation splits to reduce data leakage and improve evaluation reliability.

5.3 YOLO Training and Inference

The YOLO training process optimizes the model using the annotated training data while monitoring performance on validation data. In the project workflow, training was configured for a fixed number of epochs and the model performance was monitored throughout the run. An epoch represents one complete pass over the training dataset.

After training, the trained weights can be loaded for inference. For each input image, the model predicts candidate objects, their classes, bounding boxes and confidence values. Predictions can then be visualized by drawing the boxes and labels on the original image.

Typical Inference Flow

- Load the trained YOLO weights.
- Read an image or frame.
- Run the model on the input.
- Apply the configured confidence and non-maximum-suppression logic.
- Render bounding boxes, class names and confidence scores.
- Save or display the resulting detection.

6 Performance Test

Performance evaluation is important because an object-detection system should not be judged only by whether it can produce a prediction. The system must also be assessed for detection quality, false detections, missed objects and computational practicality.

6.1 Test Plan / Test Cases

Test Case	Input	Expected Result	Evaluation
TC-01	Image containing clearly visible crop	Crop detected with bounding box	Correct class and location
TC-02	Image containing clearly visible weed	Weed detected with bounding box	Correct class and location
TC-03	Image containing both classes	Multiple detections	Both classes localized
TC-04	Cluttered/complex vegetation	Best-effort detection	Observe false positives/negatives
TC-05	Unseen image	Generalized prediction	Qualitative generalization

6.2 Test Procedure

18. Load the final trained model weights.
19. Select representative validation or unseen images.
20. Run inference with a consistent confidence threshold.
21. Record detected classes, bounding boxes and confidence values.
22. Compare predictions against ground-truth annotations where available.
23. Review false positives, missed objects and localization errors.
24. Use quantitative metrics such as precision, recall and mAP when generated by the validation process.

6.3 Performance Outcome

The developed pipeline successfully establishes the complete path from agricultural image input to YOLO-based object detection output. The main qualitative outcome is the ability to represent crop and weed instances as localized detections instead of only assigning a single label to the complete image.

During evaluation, the most important observations are expected to come from difficult cases such as overlapping plants, small weeds, strong shadows, low contrast and background clutter. These cases should be considered when interpreting the final model metrics.

Metrics to Report from the Final Training Run

Metric	Purpose	Final Value
Precision	Measures correctness of positive detections	Insert value from final validation run
Recall	Measures how many relevant objects were detected	Insert value from final validation run
mAP@0.5	Detection accuracy using IoU threshold 0.5	Insert value from final validation run
mAP@0.5:0.95	Average precision across stricter IoU thresholds	Insert value from final validation run
Inference speed	Practical responsiveness of the detector	Measure on target hardware

Important: numerical performance values should be copied directly from the final training/validation output rather than estimated. This avoids reporting fabricated accuracy values and makes the report reproducible.

Observed Constraints

- Training time depends strongly on available CPU/GPU resources and image resolution.
- Large deep-learning dependencies require substantial storage during installation and training.
- Python package compatibility can affect the reproducibility of the environment.
- Model accuracy depends on dataset size, annotation quality and diversity of field conditions.

7 My Learnings

The project provided practical learning across the complete machine-learning development cycle. The most important learning was that successful object detection is not only about selecting a model. Data preparation, annotations, configuration, training strategy and evaluation are equally important.

Technical Learnings

- Understanding the difference between image classification and object detection.
- Understanding the basic working principle of YOLO-based detection.
- Preparing and organizing an object-detection dataset.
- Working with bounding-box annotations and class labels.
- Using the Ultralytics framework for model training and inference.
- Using Python libraries such as OpenCV, NumPy, Pandas and Matplotlib in an ML workflow.
- Monitoring training behaviour across epochs and interpreting loss/validation metrics.
- Debugging Python environments, virtual environments and package-installation problems.
- Understanding the importance of reproducible software environments.

Professional Learnings

- Breaking a large project into measurable stages.
- Documenting assumptions, problems and solutions.
- Testing a model instead of relying only on training results.
- Recognizing the difference between a prototype and a production-ready system.
- Communicating technical work through structured documentation.

8 Future Work Scope

The current project establishes a foundation for automated crop and weed detection, but a practical agricultural system would require broader datasets, stronger validation and deployment-oriented optimization.

Possible Future Improvements

25. Expand the dataset using images collected under different lighting, soil types, camera angles and crop-growth stages.
26. Increase annotation diversity and perform systematic quality checks on bounding boxes.
27. Evaluate multiple YOLO model sizes to determine the best accuracy-speed trade-off.
28. Add video-stream inference for continuous field monitoring.
29. Deploy the model on edge devices for on-site inference.
30. Integrate camera systems mounted on agricultural vehicles, drones or robots.
31. Use detection outputs to support targeted mechanical or chemical weed-management actions.
32. Develop a dashboard for visualizing detections, field coverage and model statistics.
33. Investigate domain adaptation and continual learning to handle changes between farms and seasons.
34. Perform a formal benchmark using precision, recall, mAP, latency and memory usage on the intended deployment hardware.

Conclusion

The “Crop and Weed Detection Using YOLO” project demonstrates how deep-learning-based object detection can be applied to an important precision-agriculture problem. The internship work covered dataset preparation, model training, inference and evaluation, while also providing practical experience with the challenges of real ML development environments. With additional field data and deployment-focused optimization, the system can be extended toward real-time agricultural monitoring and precision weed-management applications.

The project therefore provides a practical foundation rather than a final production system. The next stage should focus on rigorous quantitative benchmarking, larger and more diverse datasets, and deployment testing under real field conditions.